FUN-1.C.3 All local (relative) extrema occur at critical points of a function, though not all critical points are local extrema.

FUN-4.A.2 The first derivative of a function can determine the location of relative (local) extrema of the function.

5-4 Relative Extrema and the First Derivative Test

In the previous section, we were able to use $f'(x)$ to help us determine when $f(x)$ is increasing or decreasing.

In this section, we will use $f'(x)$ to determine additional information about the graph of $f(x)$.

First, a few important vocabulary terms and their technical definitions...

Definition of Extrema (Extreme Values)	
Absolute (Global) Extrema	Let $f(x)$ be defined on the closed interval $[a, b]$ with $a \leq c \leq b$. If $f(c) \geq f(x)$ for all values of x in $[a, b]$, then $f(c)$ is the **absolute maximum value** of $f(x)$ on the interval $[a, b]$. If $f(c) \leq f(x)$ for all values of x in $[a, b]$, then $f(c)$ is the **absolute minimum value** of $f(x)$ on the interval $[a, b]$.
Relative (Local) Extrema	If there is an open interval containing c on which $f(c)$ is a maximum, then $f(c)$ is called a **relative maximum**. If there is an open interval containing c on which $f(c)$ is a minimum, then $f(c)$ is called a **relative minimum**.

Let's summarize these terms:

Term	Occurs When
Relative Minimum	$f(x)$ stops decreasing and then begins to increase
Relative Maximum	$f(x)$ stops increasing and then begins to decrease
Absolute Minimum	$f(c) \leq$ all other values of $f(x)$
Absolute Maximum	$f(c) \geq$ all other values of $f(x)$

Example 1: The graph of $f(x)$ is shown below. State all relative and absolute extrema.

Abs min: $f(2) = -1$

Abs max: $f(11) = 5$

Relative min: $f(7) = 1$
$f(2) = -1$

Relative max $f(5) = 2$

Rel. max + min cannot be at endpoints

First Derivative Test (Finding Relative Extrema of $f(x)$)		
Type of Extrema	**Process**	**Explanation or Reasoning**
Relative (Local) Minimum	1. Find critical values of $f(x)$ ** $f'(x) = 0$ or undefined ** 2. Create a sign chart with $f'(x)$ 3. Check for x values where f' changes from $-$ to $+$. f' $-$ $+$	$f(x)$ has a relative minimum at $x = c$ because $f'(c) = 0$ (or $f'(c)$ is undefined) and $f'(x)$ changes sign from $-$ to $+$.
Relative (Local) Maximum	1. Find critical values of $f(x)$ ** $f'(x) = 0$ or undefined ** 2. Create a sign chart with $f'(x)$ 3. Check for x values where f' changes from $+$ to $-$. f' $+$ $-$	$f(x)$ has a relative maximum at $x = c$ because $f'(c) = 0$ (or $f'(c)$ is undefined) and $f'(x)$ changes sign from $+$ to $-$.

***Note: the function must be defined at $x = c$ to have a relative maximum or minimum.

Here's the good news: we can do ONE process to find the intervals where $f(x)$ is increasing or decreasing, and where $f(x)$ has a relative minimum or relative maximum!

Example 2: $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$. Find all relative extrema for $f(x)$. Give a reason for your answers.

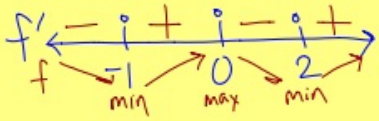
$$f'(x) = 12x^3 - 12x^2 - 24x = 0$$

$$12x(x^2 - x - 2) = 0$$

$$12x(x+1)(x-2) = 0$$

$12x = 0 \quad x+1 = 0 \quad x-2 = 0$

$x = 0 \quad x = -1 \quad x = 2$

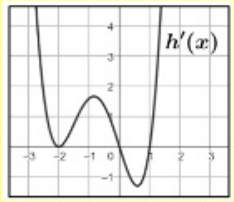


Rel. min: $f(-1) = -8$, $f(2) = -27$ because $f'(x) = 0$ at these values and $f'(x)$ changes from neg to pos at these points.

Rel max $f(0) = 5$ because $f'(0) = 0$ and $f'(x)$ changes from pos to neg at this point.

Example 4: The graph of $h'(x)$, the derivative of h, is shown at the right.

(a) On what interval(s) is $h(x)$ increasing? Give a reason for your answer.



$(-\infty, 0] \cup [1, +\infty)$

because $h'(x) \geq 0$ on these intervals.

(b) Does h have a local minimum, local maximum, or neither at $x = -2$? Give a reason for your answer.

h does not have a local max or local min at $x = -2$ because $h'(x)$ does not change signs at $x = -2$

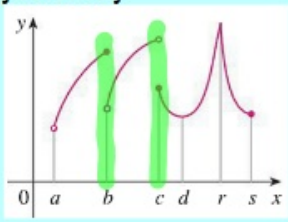
(c) Find any x values where h has a relative minimum. Explain your reasoning.

h has a rel. min at $x = 1$ because $h'(1) = 0$ and $h'(x)$ changes from neg to pos at $x = 1$

Get Ready...

This is in your notes from yesterday

For each x value below, state whether the point is a relative minimum, relative maximum, absolute minimum, absolute maximum or none of these.



a: none

b: relative max

c: none

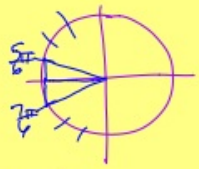
d: relative min

r: abs max, relative max

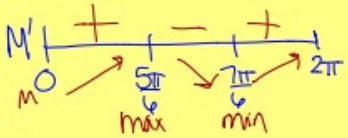
s: none

Example 6: Consider the function $M'(x)$ where the derivative of M is defined as $M'(x) = 2\cos(x) + \sqrt{3}$. Find any values of x where $M(x)$ has a local extremum on $[0,2\pi)$.

$$M'(x) = 2\cos x + \sqrt{3} = 0$$
$$2\cos x = -\sqrt{3}$$
$$\cos x = \frac{-\sqrt{3}}{2}$$
$$x = \frac{5\pi}{6} \quad x = \frac{7\pi}{6}$$



$-2+\sqrt{3}$



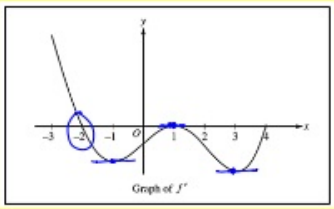
M has a local max at $x = \frac{5\pi}{6}$ because $M'(\frac{5\pi}{6}) = 0$ and M' changes from pos. to neg. at $x = \frac{5\pi}{6}$

M has a local min at $x = \frac{7\pi}{6}$ because $M'(\frac{7\pi}{6}) = 0$ and M' changes from neg. to pos at $x = \frac{7\pi}{6}$.

Example 7: (2015 AP Exam) The graph of f', the derivative of a twice-differentiable function f is shown in the figure above on the interval $[-3, 4]$. The graph of f' has horizontal tangents at $x = 1$, $x = -1$, and $x = 3$.

(a) Find all x coordinates at which f has a relative maximum. Give a reason for your answer.

$f(x)$ has a rel. max at $x=-2$ because $f'(-2)=0$ and $f'(x)$ changes from pos. to neg at $x=-2$.



(*) What are the critical values of f in the open interval $(-3, 4)$?

$x=-2$ $x=1$

because $f'(-2)=0$ & $f'(1)=0$

(**) On what open interval(s) is f decreasing? Give a reason for your answer.

$(-2,1) \cup (1,4)$ because $f'(x)<0$ on these intervals.